

## 2-1 Represent 4-Digit Numbers

**Learning goal:** I can show a 4-digit number with base-ten blocks and a place-value chart, and write it in standard form, expanded form, and word form.

**Textbook:** pages 33–36. The book models 2,138 with base-ten blocks, a place-value chart, and three written forms.

### Words to Know

|                        |                                                                                             |
|------------------------|---------------------------------------------------------------------------------------------|
| <b>digit</b>           | One of the symbols 0, 1, 2, 3, 4, 5, 6, 7, 8, 9 that we use to write numbers.               |
| <b>place value</b>     | The value a digit has because of its place. In 2,138 the 2 means 2 thousands, or 2,000.     |
| <b>base-ten blocks</b> | A <b>cube</b> is 1,000. A <b>flat</b> is 100. A <b>rod</b> is 10. A <b>unit</b> is 1.       |
| <b>standard form</b>   | The usual way to write a number with digits: <b>2,138</b> .                                 |
| <b>expanded form</b>   | The number written as the value of each digit added together: <b>2,000 + 100 + 30 + 8</b> . |
| <b>word form</b>       | The number written in words: <b>two thousand, one hundred thirty-eight</b> .                |

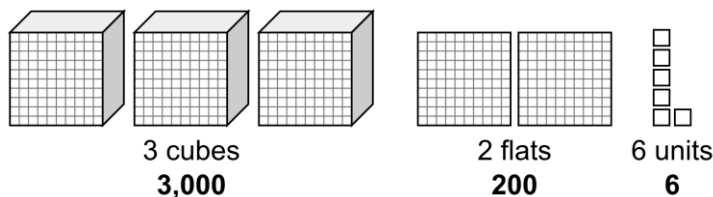
### How to Teach It

1. Build or draw the number. Count the cubes, flats, rods, and units.
2. Write each count in a place-value chart: thousands, hundreds, tens, ones. If a place has none, write **0** there.
3. Read the chart from left to right to write **standard form**. Put a comma after the thousands digit.
4. For **expanded form**, write the value of each digit and join them with + signs. Skip places that have a 0 (6,000 + 400 + 9).
5. For **word form**, say the thousands, then the word “thousand,” then read the rest like a 3-digit number. Use a hyphen for numbers like forty-one.

**Tip:** Ask, “What is this digit worth?” A 0 holds a place. In 3,206 the 0 shows there are no tens.

### Worked Examples

#### Example 1: Blocks to chart and standard form



| thousands | hundreds | tens     | ones     |
|-----------|----------|----------|----------|
| <b>3</b>  | <b>2</b> | <b>0</b> | <b>6</b> |

No rods, so write **0** in the tens place.

Standard form: **3,206**

Expanded form: **3,000 + 200 + 6**

#### Example 2: Standard form to expanded and word form

**4,152** → the digits are worth 4,000, 100, 50, and 2.

Expanded form: **4,000 + 100 + 50 + 2**

Word form: **four thousand, one hundred fifty-two**

#### Example 3: Word or expanded form to standard form

“five thousand, seventy-three”

5 thousands, 0 hundreds, 7 tens, 3 ones → **5,073**

Expanded form: **5,000 + 70 + 3**

6,000 + 400 + 9 → 6 thousands, 4 hundreds, 0 tens, 9 ones → **6,409**

### Skills Checklist *Check off each skill your child can do without help.*

- |                                                                                   |                                                                                    |
|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| <input type="checkbox"/> Tell what a cube, flat, rod, and unit are worth.         | <input type="checkbox"/> Write a number in expanded form.                          |
| <input type="checkbox"/> Write the number shown by base-ten blocks.               | <input type="checkbox"/> Read and write a number in word form.                     |
| <input type="checkbox"/> Fill in a place-value chart, using 0 for an empty place. | <input type="checkbox"/> Use place value to explain how two numbers are different. |

## 2-1 Represent 4-Digit Numbers

Name \_\_\_\_\_ Date \_\_\_\_\_

**Together** *Work with your grown-up. Talk about each step.*

- A.** What number do the base-ten blocks show? Fill in the chart. Then write the number in standard form.



| thousands | hundreds | tens | ones |
|-----------|----------|------|------|
|           |          |      |      |

Standard form: \_\_\_\_\_

- B.** Write **3,081** in expanded form and in word form.

Expanded form: \_\_\_\_\_

Word form: \_\_\_\_\_

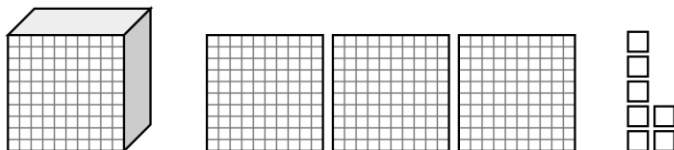
- C.** Write **four thousand, six hundred nine** in standard form and in expanded form.

Standard form: \_\_\_\_\_

Expanded form: \_\_\_\_\_

**On My Own** *Try these by yourself. Show your work.*

- 1.** What number do the base-ten blocks show? Fill in the chart.



| thousands | hundreds | tens | ones |
|-----------|----------|------|------|
|           |          |      |      |

Standard form: \_\_\_\_\_

2. Show **5,264** in the place-value chart and in expanded form.

| thousands | hundreds | tens | ones |
|-----------|----------|------|------|
|           |          |      |      |

Expanded form: \_\_\_\_\_

3. Write **eight thousand, fifty-six** in standard form and in expanded form.

Standard form: \_\_\_\_\_

Expanded form: \_\_\_\_\_

4. Write the number in standard form.

$$7,000 + 300 + 8$$

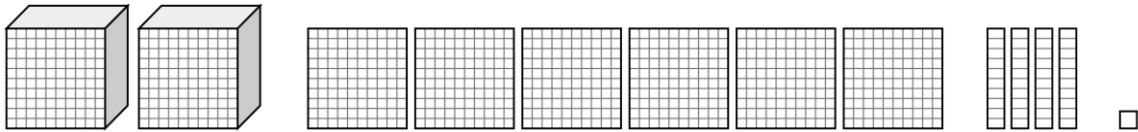
\_\_\_\_\_

5. Write the number in standard form.

$$9,000 + 60 + 4$$

\_\_\_\_\_

6. Write the number shown in standard form and in expanded form.



Standard form: \_\_\_\_\_

Expanded form: \_\_\_\_\_

7. A library had **2,374** books in May. In June it had **2,474** books. Which digit changed? Use place value to tell how many more books the library had in June.

\_\_\_\_\_  
\_\_\_\_\_  
\_\_\_\_\_

8. Use all four digits **6, 4, 9, 4** to write the 4-digit number with the **least** value. Write it in standard form, expanded form, and word form.

Standard form: \_\_\_\_\_

Expanded form: \_\_\_\_\_

Word form: \_\_\_\_\_

\_\_\_\_\_

## 2-1 Represent 4-Digit Numbers

### Together

|          |                                                                                                                                                                                      |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>A</b> | Chart: 2   4   3   5. Standard form: <b>2,435</b><br><i>2 cubes = 2,000; 4 flats = 400; 3 rods = 30; 5 units = 5.</i>                                                                |
| <b>B</b> | Expanded form: <b>3,000 + 80 + 1</b><br>Word form: <b>three thousand, eighty-one</b><br><i>There are 0 hundreds, so no hundreds part is written. Also accept 3,000 + 0 + 80 + 1.</i> |
| <b>C</b> | Standard form: <b>4,609</b><br>Expanded form: <b>4,000 + 600 + 9</b><br><i>There are 0 tens, so write 0 in the tens place. Also accept 4,000 + 600 + 0 + 9.</i>                      |

### On My Own

|          |                                                                                                                                                                                                                                                                                       |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>1</b> | Chart: 1   3   0   7. Standard form: <b>1,307</b><br><i>No rods, so the tens digit is 0. If your child writes 137, point to the empty tens place.</i>                                                                                                                                 |
| <b>2</b> | Chart: 5   2   6   4<br>Expanded form: <b>5,000 + 200 + 60 + 4</b>                                                                                                                                                                                                                    |
| <b>3</b> | Standard form: <b>8,056</b><br>Expanded form: <b>8,000 + 50 + 6</b><br><i>0 hundreds. Also accept 8,000 + 0 + 50 + 6.</i>                                                                                                                                                             |
| <b>4</b> | <b>7,308</b><br><i>No tens, so the tens digit is 0.</i>                                                                                                                                                                                                                               |
| <b>5</b> | <b>9,064</b><br><i>No hundreds, so the hundreds digit is 0.</i>                                                                                                                                                                                                                       |
| <b>6</b> | Standard form: <b>2,641</b><br>Expanded form: <b>2,000 + 600 + 40 + 1</b>                                                                                                                                                                                                             |
| <b>7</b> | The <b>hundreds</b> digit changed from 3 to 4. That is 1 more hundred, so the library had <b>100 more books</b> in June.<br><i>Accept any clear place-value explanation, e.g., "The thousands, tens, and ones are the same. Only the hundreds went up by one, so it is 100 more."</i> |
| <b>8</b> | Standard form: <b>4,469</b><br>Expanded form: <b>4,000 + 400 + 60 + 9</b><br>Word form: <b>four thousand, four hundred sixty-nine</b><br><i>Put the smallest digits in the biggest places: 4, 4, 6, 9.</i>                                                                            |